Introduction to GDB

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Preview of the Session

1. Introduction to GDB
2. GDB Commands
3. RISC-V CSR's
4. RISC-V Exceptions
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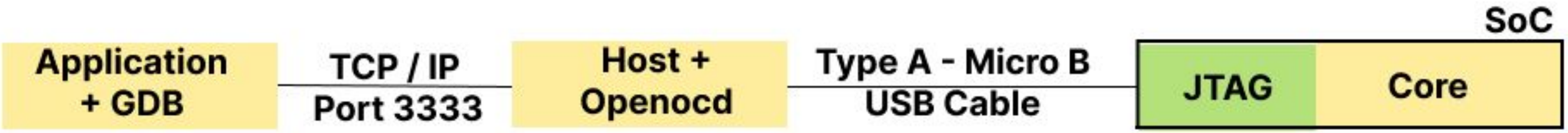
Introduction to GDB

1. GDB (GNU Debugger) -
   - Debugging tool designed for RISC V architecture.
   - Program that helps to see what the code is doing inside the processor - while it runs, stops, crashes, or behaves weirdly.



GDB Commands

Command	Functionality
set remotetimeout unlimited	Disables the timeout for that remote connection
target remote localhost : 3333	Tells GDB to connect to a remote target (e.g. OpenOCD) running on your computer at port 3333
file <path>	Informs GDB of the file with its path
load/l	Informs GDB to load the file specified in the path
compare-sections	To make sure the loaded file has been loaded properly
c/continue	To continue to execute the instruction from start

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GDB Commands

Command	Functionality
jump <*addr>	Jumps to the specified address. The debugger resumes program execution at that point unless it encounters a breakpoint there.
step / s	Steps from one function to another
stepi / si	Steps into assembler instructions, into any function calls.

GDB Commands

Command	Functionality
b break [Function Name] break [filename]:[Line Number] break [Address]	Creates a breakpoint at a specified line, address
display breakpoint / info breakpoint	To display all breakpoints
delete delete [number]	Deletes specific breakpoints or all breakpoints.

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GDB Commands

Command	Functionality
p/x <address> x/bx , x/hx , x/wx <address>	Used for displaying value of specific address "x" is hex , can be replaced with other format specifier
set {format} <address>=<value> Format (char,short,int)	Used for setting value of specific address . Format depend on the accessible size of address
info reg / i r / info registers info all-registers / i all-registers / info registers [Register name]	Displays the contents of all processor registers
bt	Backtraces to the function where exception was raised

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RISC-V Exceptions

Cause Value	Cause Meaning
0	Instruction Address Misaligned
1	Instruction Access Fault (Instruction does not exist)
2	Illegal Instruction
3	Breakpoint
4	Load Address Misaligned
5	Load Access Fault
6	Store Address Misaligned
7	Store Access Fault

RISC-V CSR's

- As part of Machine Mode Trap Handling Register RISC-V has 3 important CSR Registers

  1) mtval (machine trap value) - exception-specific information to assist software in handling the trap
  2) mepc (machine expectation pc) - points to the program counter where the exception occurred
  3) mcause (machine cause) - the value indicate the type of exception that occurred

Reference

- https://shakti.org.in/docs/RISC-V-GDB-tutorial.pdf
- https://blogshakti.org.in/risc-v-gdb-ultimate-debugging-tool-for-shakti/
- https://www.geeksforgeeks.org/linux-unix/gdb-command-in-linux-with-examples/
- https://sourceware.org/gdb/current/onlinedocs/gdb

Thank you

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